AI for Medical Imaging

Master Artificial Intelligence

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2026/09/17 – Ethics, data management and reproducibility

Important service announcement

Snellius: password reset is operational again

Next Wednesday: share last year's exam

Next Thursday (if we have time): go through the 2024's quiz

2026/09/17 – Ethics, data management and reproducibility

Data collection & management

Regulatory frameworks

(Especially in Europe)

Subject to change

Currently hotly debated

Cannot be ignored (especially in industry)

GDPR

General Data Protection Regulation (EU regulation). Requires notably:

- Patient consent
- Purpose
- Good care and handling

Retrospective studies

Require to ask consent *for that specific use*

Some exception:

- consent cannot be asked;
- “cannot be reasonably required”.

Ethical commitees

In the Netherlands (at least):

- WMO (medical research nivolving human subjects act);
- Medisch Ethische Toetsings Commissie (METC);
- probably internal one in your institute/company.

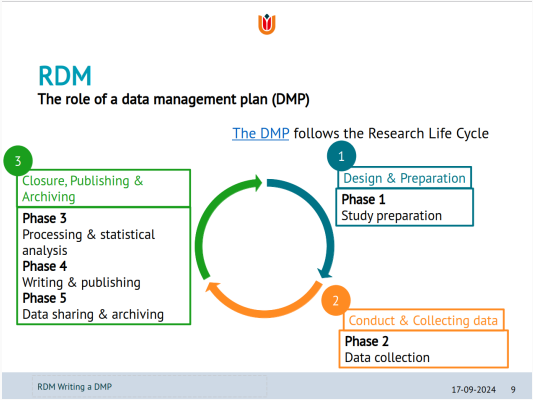
Data Management Plan (DMP) (i)

Need to lay-out *before*:

- how the security of the data will be guaranteed;
- how the quality will be guaranteed;
- what and how you will comply to the laws and regulations.

Data Management Plan (DMP) (ii)

Usually internal guidelines already exists.



Presentation at AUMC last week

Anonymous, pseudonymous

Difference:

- data: the scan, image *proper*;
- metadata: patient name, date, sex, weight, etc.

Remove personal identifiers, replace name/ID with random number:

- pseudo-anonymization: someone has the mapping, somewhere;
- anonymization: the mapping has been deleted.

But it has its limits

Example of a easy to anonymize dataset

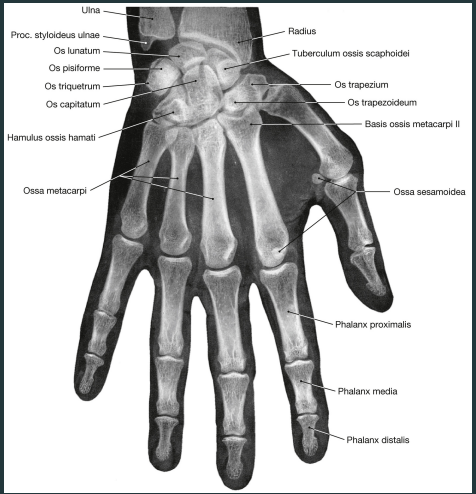
Example of face removal in MRI

On patient “uniqueness”: side read

The end of average

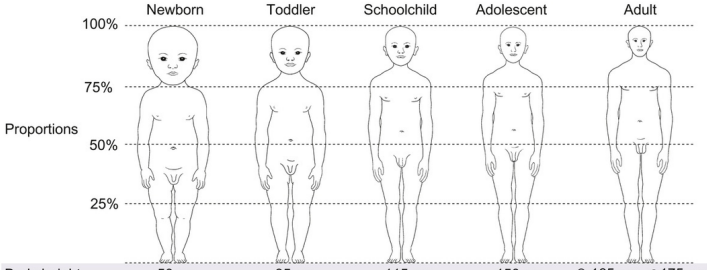
Todd Rose, 2016, ISBN-10: 9780141980034, https://zoeken.oba.nl/detail/Todd-Rose/De-mythe-van-het-gemiddelde-succesvol-zijn/Boek/?itemid=%7Coba-catalogus%7C1156761

Live experiment



Finger length as finger-printing technique

Kids: not just “small” adults

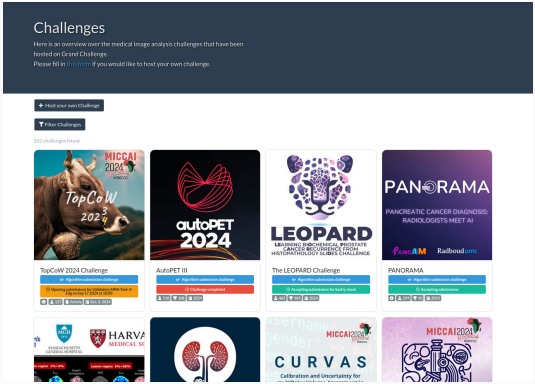


	Newborn	Toddler	Schoolchild	Adolescent	Adult
Body height	50 cm	85 cm	115 cm	150 cm	♀ 165 cm ♂ 175 cm
Body weight	3,3 kg	12 kg	20 kg	40 kg	♀ 60 kg ♂ 67 kg
Heart rate	140/min	120/min	95/min	85/min	70/min
Blood pressure	60/35 mmHg	90/50 mmHg	100/60 mmHg	110/70 mmHg	120/80 mmHg
Respiration rate	40/min	25/min	20/min	17/min	15/min

Source: Sobotta Atlas of Anatomy, 17th edition

Licensing

Check the license carefully, e.g.: https://creativecommons.org/licenses/by/4.0/ or https://creativecommons.org/licenses/by-nc/4.0/



https://grand-challenge.org/challenges/

On collaboration and teamwork

If not within the same group, often require a *Data Transfert Agreement (DTA)*

Not everyone in the project has access to the data

People have different expertise, and focus

Code storage

Many options:

- Use facilities from company (private Gitlab, hosted on facility)
- Local git repo somewhat backed up
- Public/private Github *without any patient data saved there*

Data breach

For security: the best data is the data you do not have

Limit the scope of what you collect

In case of breach/leak: *your institution* report it as soon as possible

Electronic hygiene

Always valid, but *especially* if you handle sensitive data:

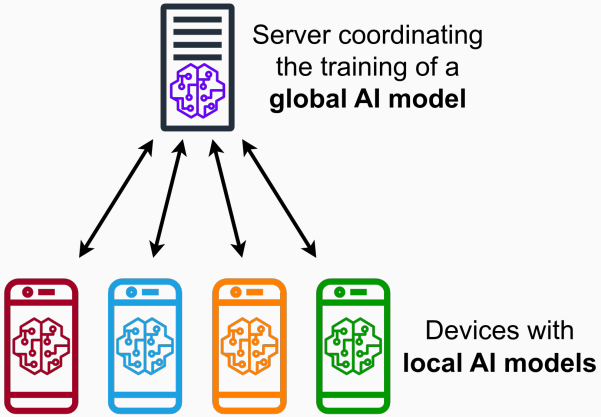
- Use strong passphrases for the ones you need to remember (ex.: `https://github.com/ligos/readablepassphrasegenerator`),
- and password manager for the rest;
- use a 2nd factor (physical token are great);
- encrypt your drives;
- *do not* plug a “found” USB key in your computer.

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Trained models

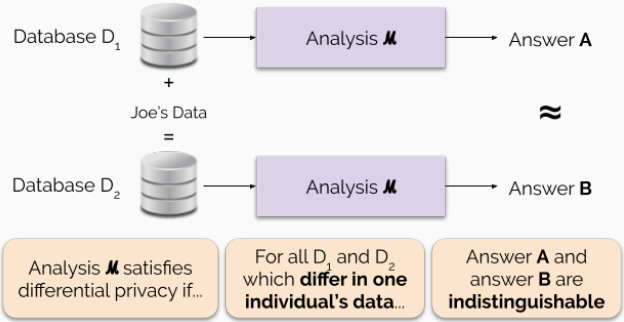
Federated learning

When data cannot be shared even for training



https://en.wikipedia.org/wiki/Federated_learning

Privacy-preserving learning



Key word: https://en.wikipedia.org/wiki/Differential_privacy

Drawbacks

No free lunch

Both federated learning and differential privacy:

- decreased performances;
- higher-computational cost.

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Reproducibility

For papers

`https://www.midl.io/reproducibility`:

- Is the reproducibility of the work addressed?
- Is the code publicly available?
- Are public datasets used?
- Are the required package dependencies listed?
- Is the code for model training available?
- Is the code for model evaluation available?
- Is the presented trained model available?
- Is there documentation for the available material?
- Is there licensing for the available material?

But what does *reproducible* mean

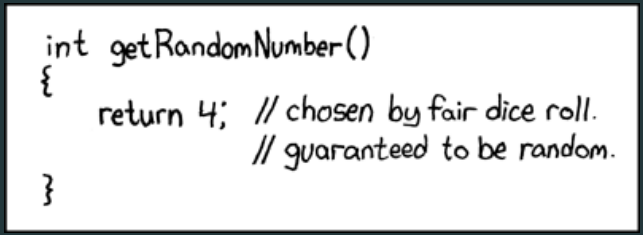
- Are the trained network parameters bit-wise identical;
- can you train the same model (code + data) on a different machine;
- or with slightly package versions;
- are results within a defined confidence interval (which one)?

The numerous sources of randomness

We use *stochastic* (i.e., random) gradient descent. Notably:

- network weights are initialized randomly;
- batch sampling is random;
- default float operation on GPU are non-deterministic;
- hardware architecture might behave *slightly* differently;
- kernel versions might have some variation.

Not covered: software, compiler and hardware bugs.



https://xkcd.com/221/

True randomness vs pseudo-randomness

True randomness is very difficult to get on a computer.

Only truly required for cryptography

Concrete example:
https://www.cloudflare.com/learning/ssl/lava-lamp-encryption/

Alternative: *pseudo*-random

Appears random, in fact generated by deterministic process, from one "seed".

(Linux:) `cat /proc/sys/kernel/random/entropy_avail`

Eliminating some sources of (pseudo-)randomness

Setting the seed:

```
import random
random.seed(0)
import numpy as np
np.random.seed(0)
import torch
torch.manual_seed(0)
```

Using deterministic algorithms: https://pytorch.org/docs/stable/generated/torch.use_deterministic_algorithms.html

Are you really testing a model, a seed, an “artifact”?

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Inference, deployment, and responsibility

“Move fast and break things”

— Discontinued motto at Facebook [now Meta]

On engineering and responsibility

Many other fields of engineering: responsible for your design/construction



https://en.wikipedia.org/wiki/Iron_Ring

Somehow software "engineering" escapes this

`https://en.wikipedia.org/wiki/Therac-25`

“People should stop training radiologists now. It’s just completely obvious that within five years deep learning is going to do better than radiologists.” — Geoffrey Hinton (2016)

“Should radiologists be worried about their jobs? Breaking news: We can now diagnose pneumonia from chest X-rays better than radiologists.” — Andrew Ng (2017)

Working with, or against, the clinical doctors

Current [unproved] consensus: radiologists using “AI” will do better than the ones without, and “AI” will not replace them

- Could accelerate some tasks
- Could reduce the need of second opinion
- Can help to make some grading more consistent

Tough legal nut: who is responsible

If using some predictive system: who is ultimately responsible?

- the ones collecting the data;
- the engineers designing and training the model;
- the company selling it;
- the regulatory body that approved it;
- the the one using it?

Lessons from aerospace (personal take)

Automation can lead to loss of intuition and understanding of the system

https://en.wikipedia.org/wiki/Air_France_Flight_447

https://admiralcloudberg.medium.com/
the-long-way-down-the-crash-of-air-france-flight-447-8a7678c37982

Confirmation bias is another psychological risk to mitigate:

https://en.wikipedia.org/wiki/Air_Transat_Flight_236

https://admiralcloudberg.medium.com/
crisis-over-the-atlantic-the-near-crash-of-air-transat-flight-236-671d3a0

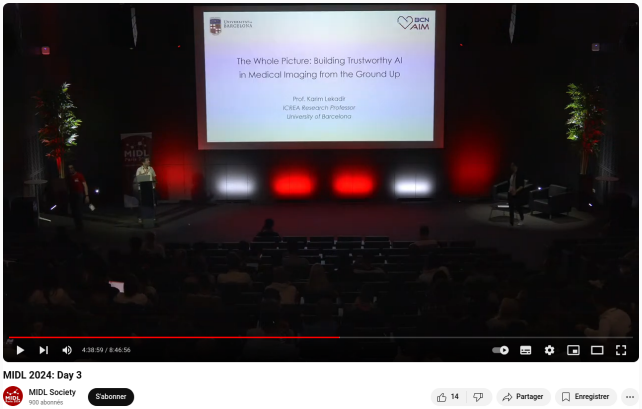
Other considerations

- Fairness
- Robustness
- Interpretability

Keynote at MIDL 2024

Karim Lekadir

The whole picture: Building trustworthy AI in medical imaging from the ground up



https://www.youtube.com/live/-mV53dZZs9o?feature=shared&t=16739

Keynote at MIDL 2025

Mirabela Rusu

Multimodal learning for early cancer detection in low resource settings (and a few lessons learned along the way)



https://www.youtube.com/live/GYrQgWC1peE?si=FThwAgplcnqtU4Vn&t=9400

Recommended after last lecture on image registration.